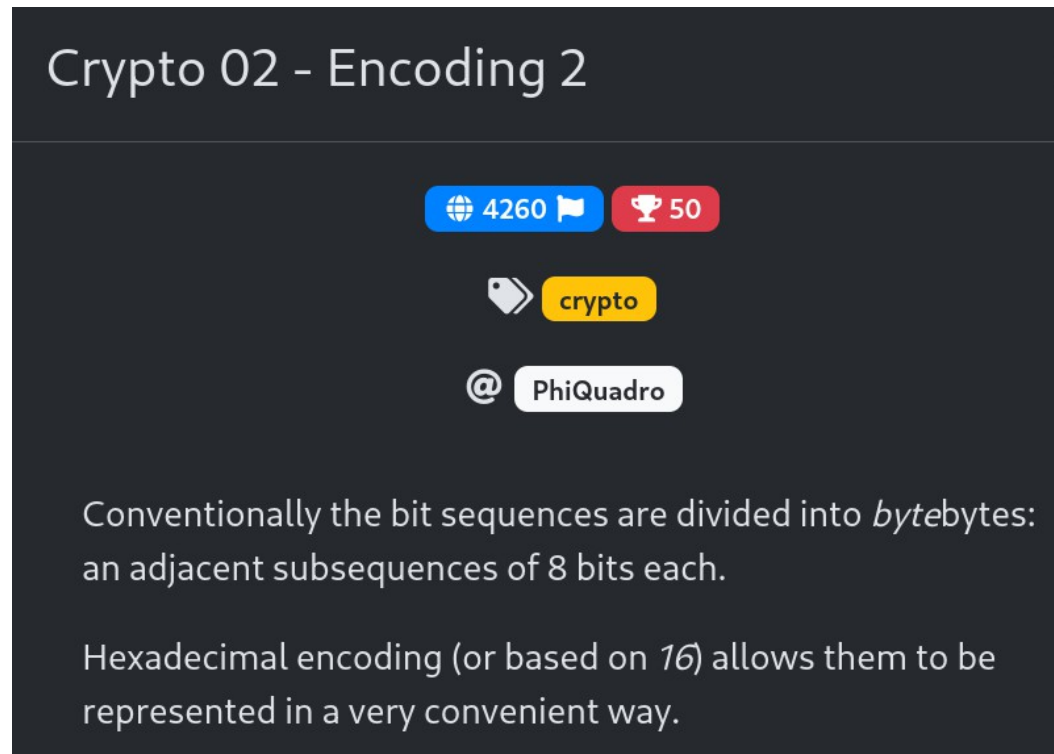


OLlcyber Introduction

Crypto 02 – Encoding 2



In this challenge, we're asked to decode hexadecimal strings into bytes

Crypto 02 – Encoding 2

What are Hexadecimal Strings?

Hexadecimal 41 42 43 44 →
ASCII ABCD

A hexadecimal (hex) string is a number of pairs of hex digits, which can be encoded into bytes, which can be rendered as ASCII characters

Crypto 02 – Encoding 2

Hexadecimal String to ASCII

To get the flag, turn the following string into the corresponding bytes sequence.

```
666c61677b68337834646563696d616c5f63346e5f62335f414  
24144424142457d
```

This challenge requires us to convert a hexadecimal string to a byte sequence, in this case, ASCII characters

Crypto 02 – Encoding 2

CyberChef Solve

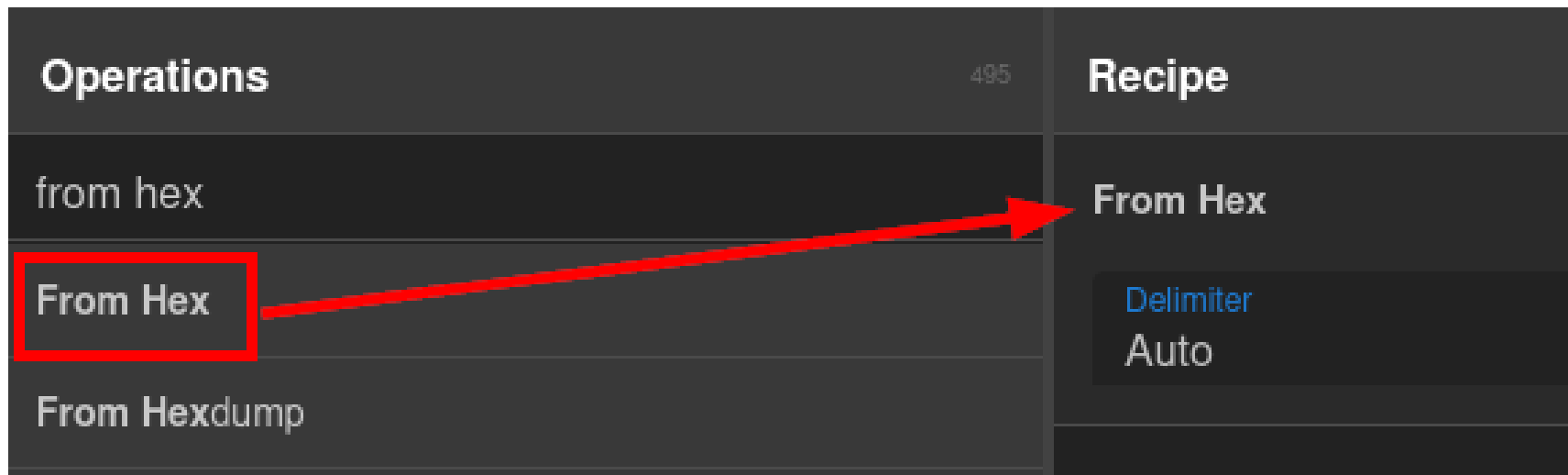
Input

```
666c61677b68337834646563696d616c5f63
```

To solve this challenge in CyberChef, first we need to copy the hexadecimal string, then paste it into the `Input` window

Crypto 02 – Encoding 2

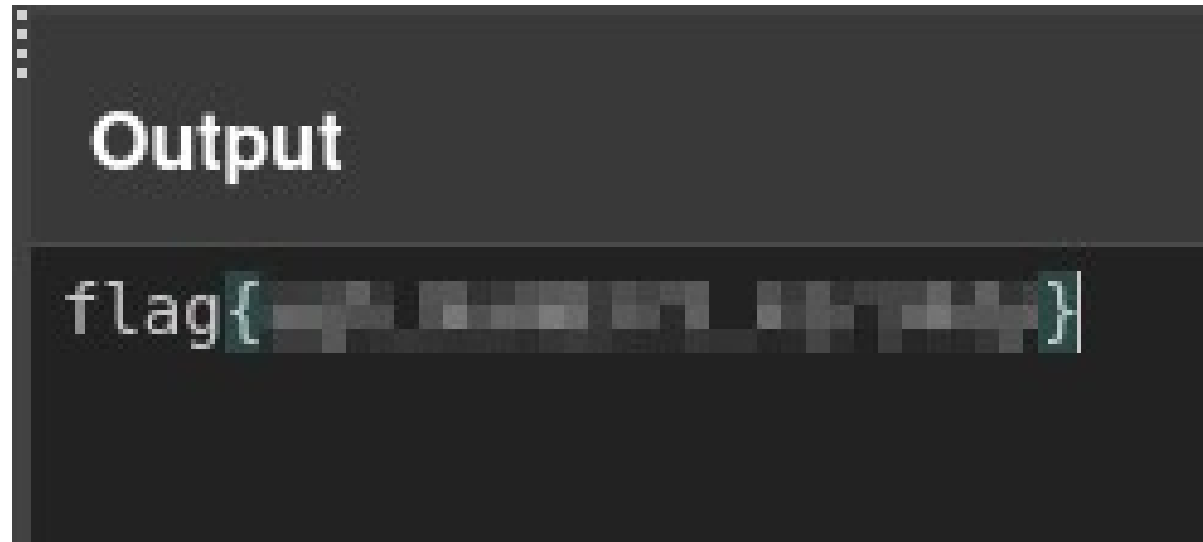
CyberChef Solve



Then, in the `Operations` window, search for the `From Decimal` operation, then drag that selection into the `Recipe` window

Crypto 02 – Encoding 2

CyberChef Solve

A screenshot of a dark-themed window titled "Output". Inside the window, the text "flag{" is visible, followed by a series of blurred, pixelated characters, and then a closing brace "}". The text is white and appears to be a flag from a challenge.

```
Output
```

```
flag{[blurred characters]}
```

The operation will convert the hexadecimal numbers to their ASCII character equivalents, which we can see in the `Output` window

Crypto 02 – Encoding 2

Linux Solve

```
echo '666c6...' | xxd -r -p
```

We can solve this challenge using Linux commands by using the `echo` and `xxd` commands

Crypto 02 – Encoding 2

Linux Solve

```
echo '666c6...' | xxd -r -p
```

First, the `echo` command is used to feed the hexadecimal into the next command

Crypto 02 – Encoding 2

Linux Solve

```
echo '666c6...' | xxd -r -p
```

Then the `xxd` command is used to convert the hexadecimal string to a printable string

Crypto 02 – Encoding 2

Python Solve

```
hex_string = '666c61677b6833783464656369'

flag_bytes = bytes.fromhex(hex_string)

print(flag_bytes)
```

To solve this challenge with Python, we can use the hexadecimal string as a string variable

Crypto 02 – Encoding 2

Python Solve

```
hex_string = '666c61677b6833783464656369'

flag_bytes = bytes.fromhex(hex_string)

print(flag_bytes)
```

Then define another variable, running the `bytes.fromhex` method to convert the hex string into its bytes equivalent

Crypto 02 – Encoding 2

Python Solve

```
hex_string = '666c61677b6833783464656369'

flag_bytes = bytes.fromhex(hex_string)

print(flag_bytes)
```

Finally, we print out those bytes, which prints out with no problems, since all of the bytes are printable (not all bytes are printable)